

1: [35 points total] You are interested in the effect of a binary treatment on an outcome of interest. To this end, you collect an iid sample of n individuals and assign them to the treatment or control group with equal probability. Let Y_i denote the outcome of the i th individual. It is assumed that $E[Y^2] < \infty$. Let D_i denote the treatment status of the i th individual, where $D_i = 1$ if the i th individual is treated and $D_i = 0$ if the i th individual is not treated. You assume the following model for Y_i :

$$Y_i = \alpha_i + \beta_i D_i$$

where (α_i, β_i) are independent of D_i . Note that α_i and β_i are random variables, and β_i is the effect of the treatment on the outcome. Furthermore the effect is "heterogeneous" in the sense that β_i differs across individuals.

1. [5 points] Explain why it might be reasonable to assume that (α_i, β_i) is independent of D_i .

Each individual is assigned with equal probability to the treatment and control group, therefore the individual characteristics that determine alpha and beta are independent of D_i .

2. [10 points] Consider the linear regression:

$$Y_i = \alpha + \beta D_i + \varepsilon_i$$

which you interpret as the best linear predictor of Y_i given D_i . Express α and β in terms of the distribution of (α_i, β_i) . In particular, show that $\beta = E[\beta_i]$

$$\beta = \frac{\text{Cov}(Y_i, D_i)}{\text{Var}(D_i)} = \frac{\text{Cov}(\alpha_i + \beta_i D_i, D_i)}{\text{Var}(D_i)} = \frac{\text{Cov}(\alpha_i, D_i)}{\text{Var}(D_i)} + \frac{\text{Cov}(\beta_i D_i, D_i)}{\text{Var}(D_i)} = \frac{\text{Cov}(\beta_i D_i, D_i)}{\text{Var}(D_i)}$$

where the last equality follows (a)'s implication that $\text{Cov}(\alpha_i, D_i) = 0$.

$$\text{Cov}(\beta_i D_i, D_i) = E[\beta_i D_i^2] - E[\beta_i D_i] E[D_i] = E[\beta_i] (E[D_i^2] - E[D_i]^2) = E[\beta_i] (\text{Var}[D_i])$$

So $\beta = \frac{E[\beta_i] (\text{Var}[D_i])}{\text{Var}(D_i)} = E[\beta_i]$. Then,

$$\alpha = E[Y_i] - \beta E[D_i] = E[\alpha_i + \beta_i D_i] - \beta E[D_i] = E[\alpha_i] + E[\beta_i D_i] - \beta E[D_i]$$

and since β_i is independent of D_i

$$= E[\alpha_i] + E[\beta_i] E[D_i] - \beta E[D_i] = E[\alpha_i] + \beta E[D_i] - \beta E[D_i] = E[\alpha_i]$$

3. [5 points] Do you think ε_i is heteroskedastic or homoskedastic? Why?

(hint: try writing $\varepsilon_i = \alpha_i - E[\alpha_i] + (\beta_i - E[\beta_i]) D_i$ and investigating $\text{Var } \varepsilon_i | D_i$ for both possible values of D_i .)

Following the hint:

$$\begin{aligned}\text{Var } \varepsilon_i | D_i &= \text{Var} [\alpha_i - E[\alpha_i] + (\beta_i - E[\beta_i]) D_i | D_i] \\ &= \text{Var} [\alpha_i + \beta_i D_i - E[\beta_i] D_i | D_i]\end{aligned}$$

which equals $\text{Var} [\alpha_i]$ if $D_i = 0$ and equals $\text{Var} [\alpha_i + \beta_i]$ if $D_i = 1$. Since these are not generally equivalent, the error term ε_i is most likely heteroskedastic.

4. [5 points] Define $X_i = (1, D_i)$ and define $\hat{\alpha}_n, \hat{\beta}_n$ as the OLS estimators of α and β given our sample. Recall from our usual asymptotic theory that we know:

$$\sqrt{n} \left((\hat{\alpha}_n, \hat{\beta}_n)' - (\alpha, \beta)' \right) \xrightarrow{d} N(0, \Omega)$$

Write down an expression for Ω as well as a consistent estimator of Ω . Call this estimator $\hat{\Omega}_n$. You do not need to show that $\hat{\Omega}_n$ is consistent.

From the previous question, we know we must use heteroskedasticity robust standard errors. Therefore, we have:

$$\Omega = E[X_i X_i']^{-1} \text{Var}[X_i \varepsilon_i] E[X_i X_i']^{-1}$$

And:

$$\hat{\Omega}_n = \left(\frac{1}{n} \sum_{i=1}^n X_i X_i' \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n X_i X_i' \varepsilon_i^2 \right) \left(\frac{1}{n} \sum_{i=1}^n X_i X_i' \right)^{-1}$$

5. [10 points] Suppose we are interested in obtaining a confidence interval around $\hat{\beta}_n$, but we do not care about $\hat{\alpha}_n$. Define $r' = (0, 1)'$ and show how to use this vector to derive a confidence interval for $\hat{\beta}_n$ at level $a \in (0, 1)$.

By the Continuous Mapping Theorem,

$$r' \hat{\Omega}_n r \xrightarrow{p} r' \Omega r$$

Also by the Continuous Mapping Theorem,

$$\begin{aligned}\sqrt{n} \left(r' (\hat{\alpha}_n, \hat{\beta}_n)' - r' (\alpha, \beta)' \right) &\xrightarrow{d} N(0, r' \Omega r) \\ \Rightarrow \sqrt{n} (\hat{\beta}_n - \beta) &\xrightarrow{d} N(0, r' \Omega r)\end{aligned}$$

Slutsky's and CMT gives us

$$\frac{\sqrt{n} \left(\hat{\beta}_n - \beta \right)}{\sqrt{r' \hat{\Omega}_n r}} \xrightarrow{d} N(0, 1)$$

Therefore we make our confidence region:

$$C_n := \left\{ c \in R : \frac{\sqrt{n} \left| \hat{\beta}_n - c \right|}{\sqrt{r' \hat{\Omega}_n r}} \leq z_{1-a/2} \right\}$$

where $z_{1-a/2}$ is the $1 - a/2$ quantile of standard normal distribution.